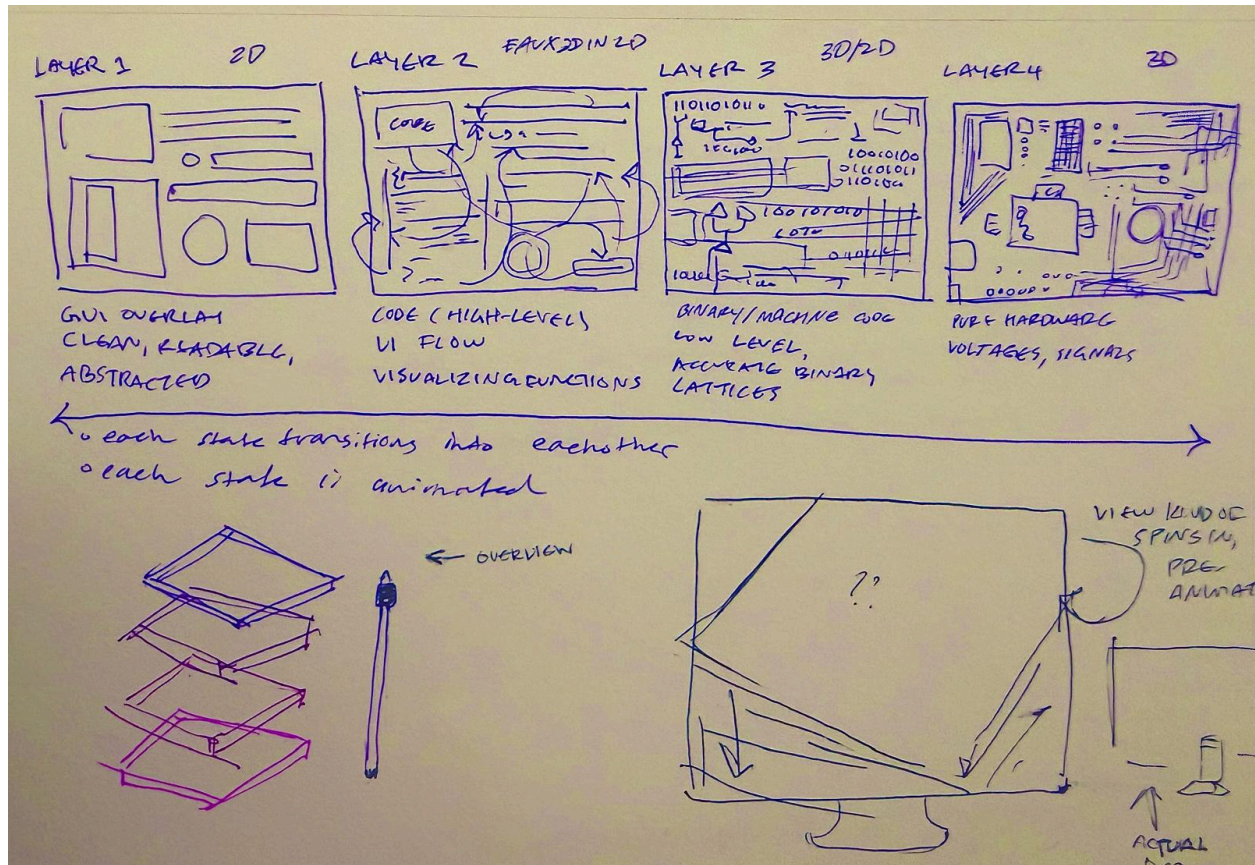


Research & Development, Pt. 2

Midterm Project: Abstraction

1. Drawing

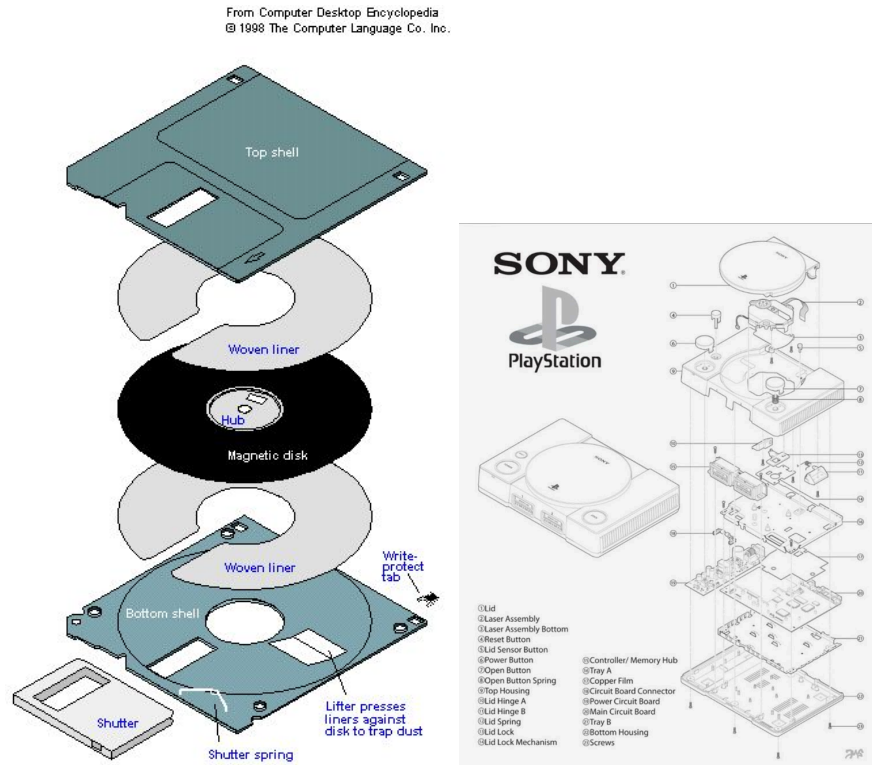


2. References

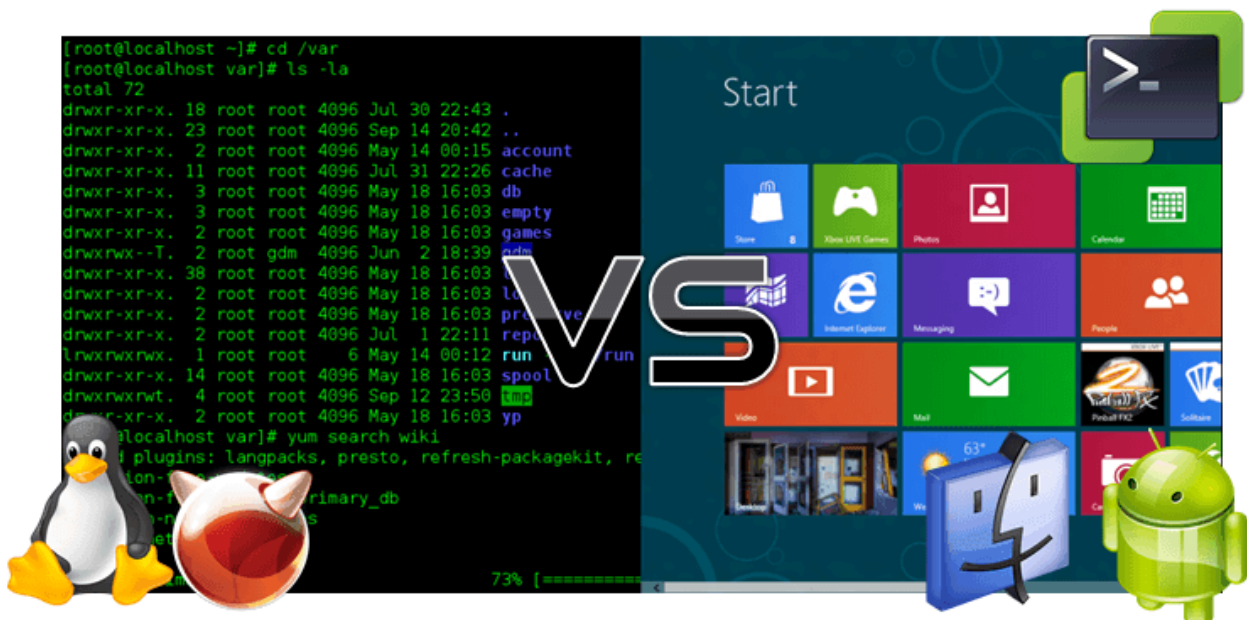
<https://animejs.com>

I'm interested in how anime.js demonstrates smooth, continuous motion driven by a single parameter over time. The animations feel precise and controlled, and transitions are gradual rather than abrupt. This is relevant to my project because abstraction is not presented as a series of cuts, but as a continuous transformation. I want the movement between layers to feel fluid and parameter-based rather than segmented. Achieving this will require careful control of interpolation, timing, and easing within TouchDesigner when mapping the slider to the animation timeline.

<https://pin.it/7k1rlhnWu> (similar to [anime.js](https://pin.it/7k1rlhnWu) but much more intense/stylized, good visual ref)



The exploded hardware diagrams are what I'm looking for because they separate components spatially while preserving structural relationships. The viewer can understand how parts depend on one another even when visually divided. This directly informs how I plan to present abstraction layers: vertically separated but clearly connected; The clarity in these diagrams is achieved through controlled spacing, consistent perspective, and minimal color variation. To produce a similar effect, I will need precise modeling in Blender, clean lighting, and restrained material design.



4. Prospective Physical Components (i have most of the stuff)

- ☐ 27" (minimum) monitor for display (rent from rc, email lisa)
- ☒ Laptop running TouchDesigner
- ☒ ~~Blender-rendered video file~~
- ☒ microcontroller
- ☐ Linear potentiometer (physical slider) (buy) -> NEW: Use a potentiometer or dial
- ☒ ~~USB cable (serial coms)/Power supply and extension cables~~
- ☐ Fabricated enclosure for slider (3D printed or laser cut) (gotta print) -> NEW: i think i could also use carefully cut cardboard because idk if a single potentiometer needs a whole PLA case
- ☐ Table or pedestal for presentation (can probably find one or use a table somewhere. But have 2 check.)

5. Technical Deets & Feasibility

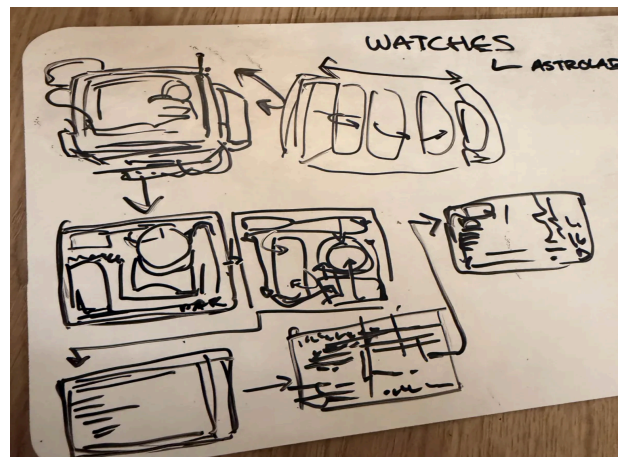
Interaction design:

- Viewer moves slider.
- Slider maps to animation timeline.
- Animation eases gradually.
- Each layer transitions with easing & vfx.
- Viewer can move freely but pacing is visually controlled.

To ensure feasibility within the timeline, the abstraction sequence will be pre-rendered in Blender as a single continuous animation representing four distinct layers: hardware, binary/machine representation, high-level code, and graphical interface. Rather than constructing fully interactive 3D environments, TouchDesigner will be used to map a physical slider input to the animation timeline, allowing viewers to scrub through abstraction levels in real time.

[[Skills (crossing off stuff i 100% know how to do)]]

- ☒ ~~Blender modeling~~
- ☐ Blender animation (i've done some but not to this extent...i'll figure it out)
- ☒ ~~Lighting and material control for layered differentiation~~
- ☒ ~~After effects/nuke for compositing~~
- ☒ TouchDesigner timeline mapping
- ☒ Serial input mapping (Arduino to TouchDesigner)
- ☒ ~~GHOP smoothing and value normalization~~



Concept para:

This project visualizes the layers of abstraction that exist within digital computing systems. Although users interact primarily with graphical interfaces, those interfaces sit on top of multiple hidden layers including high-level code, machine instructions, and physical hardware, resulting in less cognitive load at the cost of deep understanding and complexity. The piece reveals these layers by gradually transitioning between them through a continuous animation. Instead of presenting each layer separately, the viewer scrubs through a single transformation where each level emerges from the previous one.

Presentation plan:

The piece will be presented as an interactive installation. A monitor will display the animation while a physical slider placed on a pedestal allows viewers to control the level of abstraction. Moving the slider scrubs through the animation timeline, revealing transitions between hardware, machine representation, code, and graphical interface.

The slider will be connected to a microcontroller that sends serial input to TouchDesigner, which maps the value to the animation timeline. Wiring will be concealed inside a fabricated enclosure and routed beneath the pedestal to avoid distracting from the visual experience. The physical slider reinforces the concept of navigating abstraction layers by making the viewer actively control the transition between them.

Connection:

The project expands the conventional use of screen-based media. Instead of passively watching a video, viewers control the animation's position using a physical device, effectively navigating through the conceptual structure of computing systems. The work also examines how digital technology hides its underlying complexity. Most users interact only with graphical interfaces, while the lower layers of computation remain invisible. Abstraction makes systems easier to use and understand, but that simplification can erase nuance. Information that exists in lower layers becomes compressed into higher-level representations that appear clean and intuitive. The project visualizes this tradeoff, showing that simplicity is often produced by hiding the complexity that makes it possible.